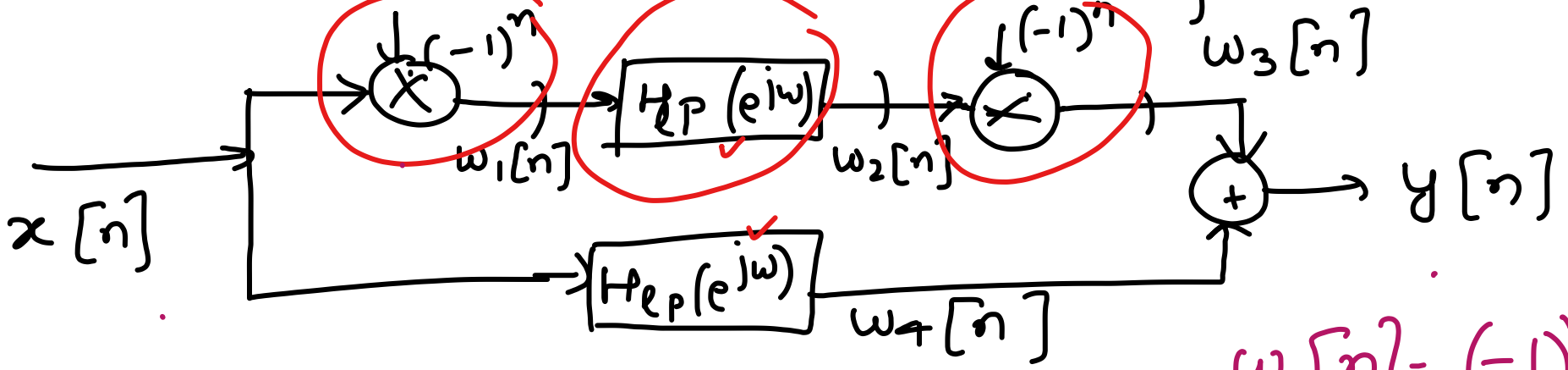
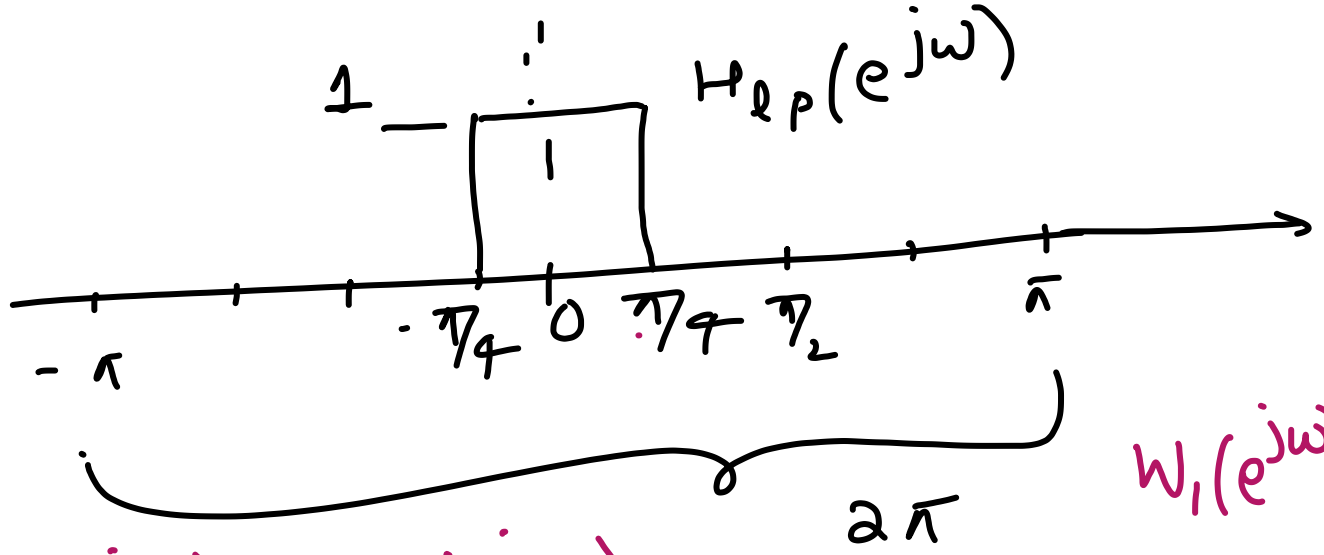
Nov. 24, 2025

Tutorial & Problem solving





$$w_1[n] = (-1)^n x[n]$$

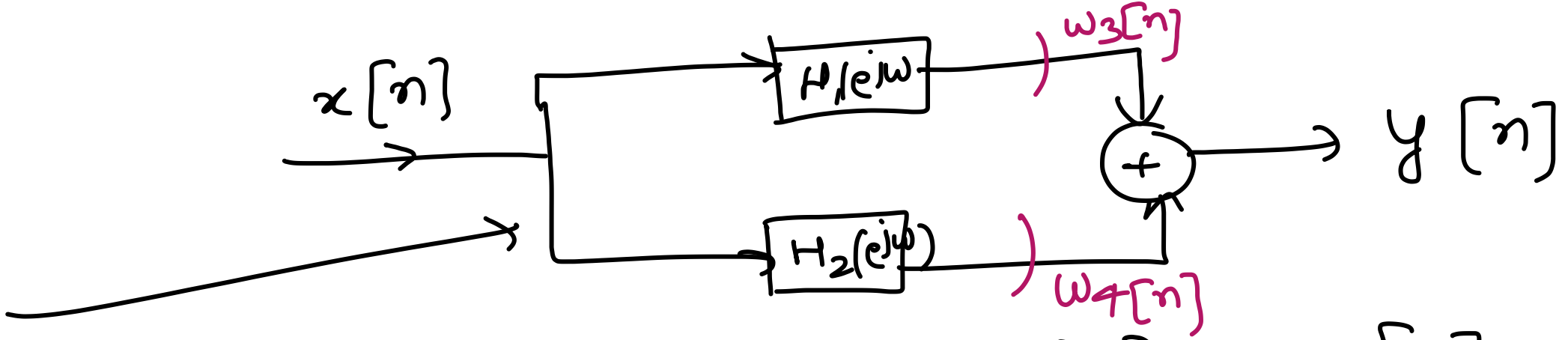
$$= e^{j\pi n} x[n]$$

!! frequency shifting property

$$W_1(e^{j\omega}) = X(e^{j(\omega-\pi)})$$

$$W_2(e^{j\omega}) = W_1(e^{j\omega}) H_{lp}(e^{j\omega})$$

$$W_3(e^{j\omega}) = W_2(e^{j(\omega-\pi)})$$



Systems are in parallel.

$$y[n] = x[n] * h_1[n] + x[n] * h_2[n]$$

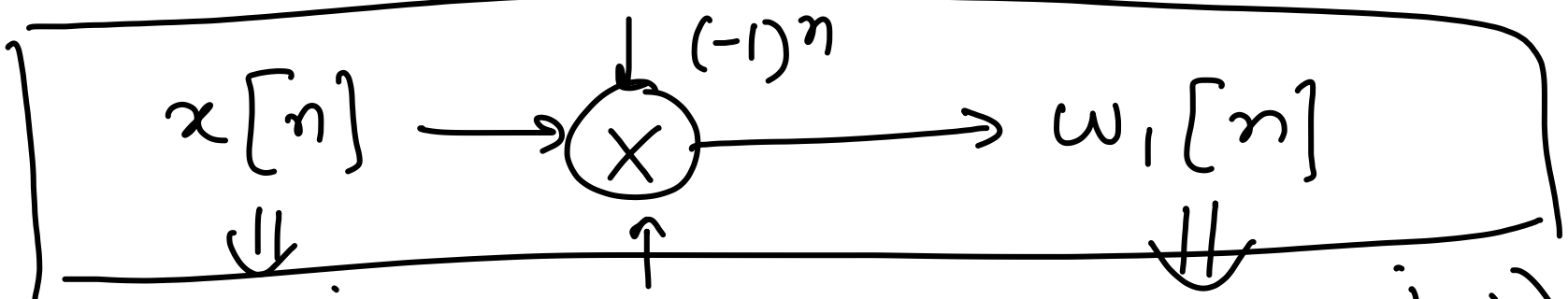
$$= x[n] * \{ h_1[n] + h_2[n] \}$$

$$H(e^{j\omega}) = \mathcal{F}\{ h[n] \}$$

$$= \mathcal{F}\{ h_1[n] + h_2[n] \}$$

$$H(e^{j\omega}) = H_1(e^{j\omega}) + H_2(e^{j\omega})$$

$$= H_1(e^{j\omega}) + H_{ep}(e^{j\omega})$$



$$W_1(e^{j\omega}) = X(e^{j(\omega-\pi)})$$

Proof:

$$w_1[n] = x[n]\,(-1)^n$$

$$= e^{j\pi n}\, x[n] =$$

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$? \rightleftharpoons X(e^{j(\omega-\omega_0)}) = G(e^{j\omega})$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$g[n] = \frac{1}{2\pi}\int_{2\pi} G(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{2\pi} X(e^{j(\omega-\omega_0)})\, e^{j\omega n}\, d\omega$$

Let $\omega - \omega_0 = \omega'$

$d\omega = d\omega'$

$\omega = \omega' + \omega_0$

$$= \frac{1}{2\pi} \int_{2\pi} X\left(e^{j\omega'}\right) e^{j(\omega'+\omega_0)n} \, d\omega'$$

$$g[n] = e^{j\omega_0 n} \cdot \frac{1}{2\pi} \int_{2\pi} X\left(e^{j\omega'}\right) e^{j\omega' n} \, d\omega'$$

$$= e^{j\omega_0 n} \, x[n]$$

if $\omega_0 = \pi$

$$g[n] = e^{j\pi n} \, x[n]$$

$$G(e^{j\omega}) = X\left(e^{j(\omega-\pi)}\right) \times$$

$$W_1(e^{j\omega}) = X\left(e^{j(\omega-\pi)}\right)$$

$$W_2(e^{j\omega}) = W_1(e^{j\omega})\, H_{ep}(e^{j\omega})$$

$$W_3(e^{j\omega}) = W_2\left(e^{j(\omega-\pi)}\right)$$

$$= W_1\left(e^{j(\omega-\pi)}\right) H_{ep}\left(e^{j(\omega-\pi)}\right)$$

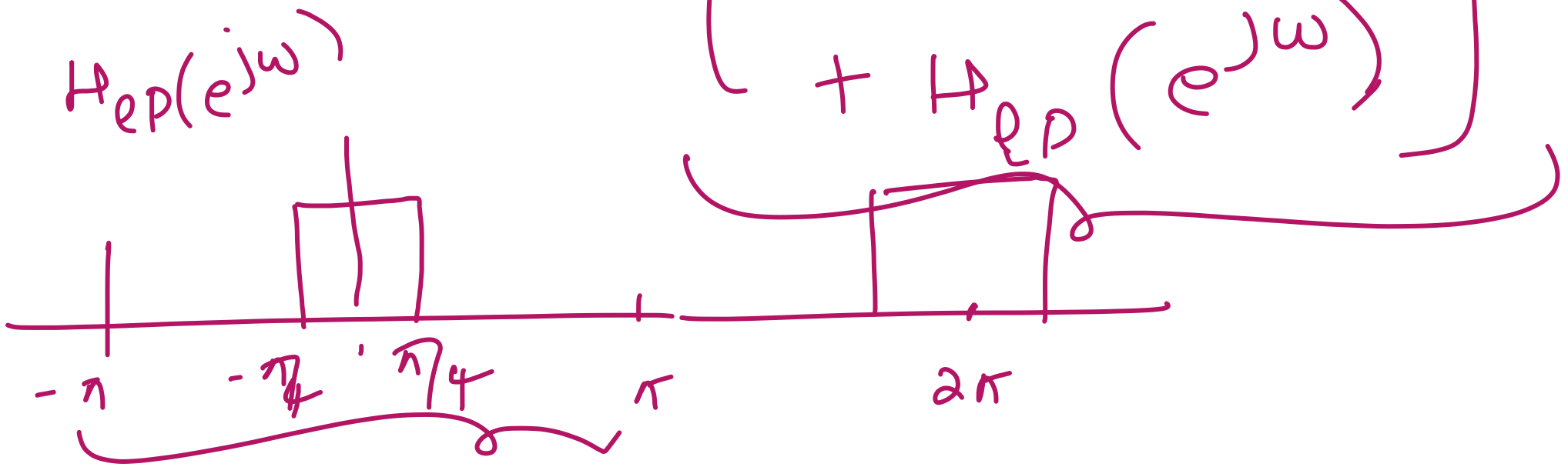
$$= X\left(e^{j(\omega-2\pi)}\right) H_{ep}\left(e^{j(\omega-\pi)}\right)$$

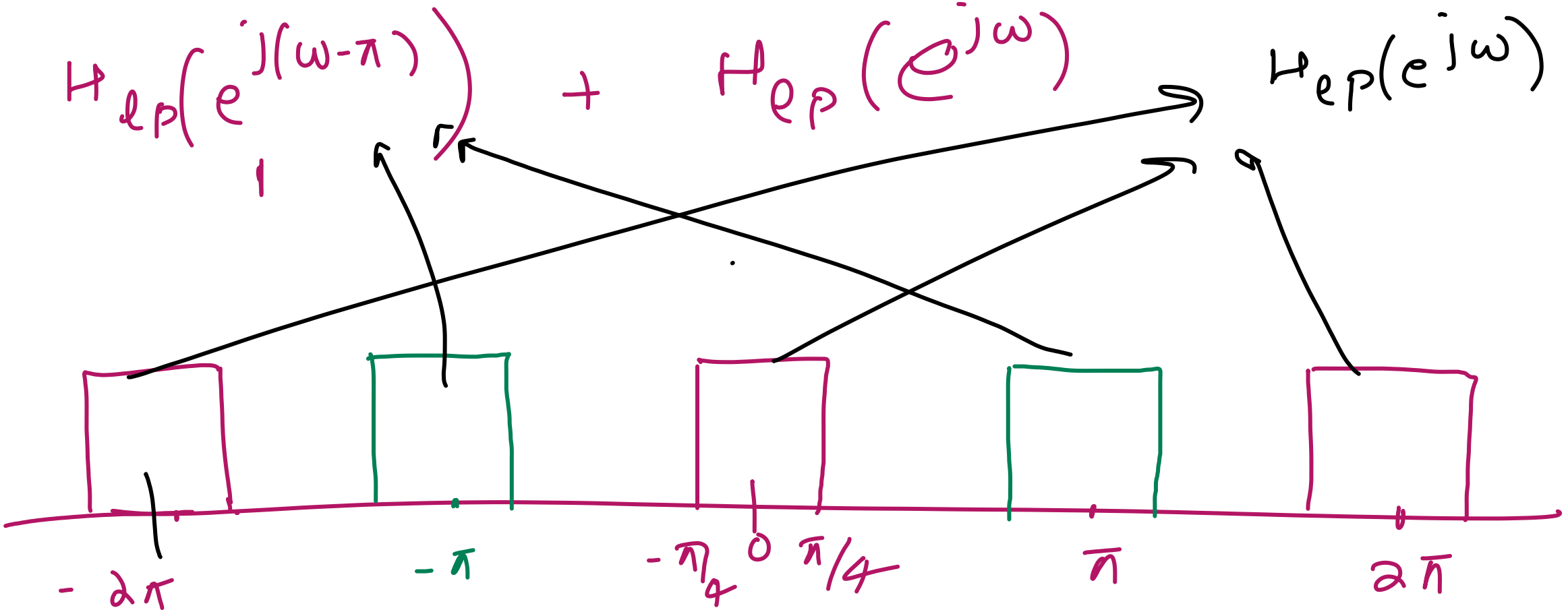
$$W_3(e^{j\omega}) = X(e^{j\omega})\, H_{lp}\left(e^{j(\omega-\pi)}\right)$$

$$\Downarrow$$

$$H_1(e^{j\omega})$$

$$Y(e^{j\omega}) = X(e^{j\omega})\left[H_{lp}(e^{j(\omega-\pi)}) + H_{lp}(e^{j\omega})\right]$$





↑ overall system

9.10 $H_1(s) = \dfrac{1}{(s+1)(s+3)}$, $\mathrm{Re}\{s\} > -1$

At $s = j\omega \Longrightarrow$ F.T. $\longrightarrow$ Lowpass

$$H(j\omega) = \frac{1}{(j\omega+1)(j\omega+3)}$$

as $\omega \to 0 \Rightarrow |H(j\omega)| = \frac{1}{3}$

as $\omega \to \infty \Rightarrow |H(j\omega)| \Rightarrow 0$

$$H(j\omega) = \frac{1}{-\omega^2 + 4j\omega + 3}$$

$$= \frac{1/4}{j\omega + (3-\omega^2)/4}$$

$$= \frac{A}{j\omega + B}$$

$$= \frac{A(-j\omega + B)}{\omega^2 + B^2}$$

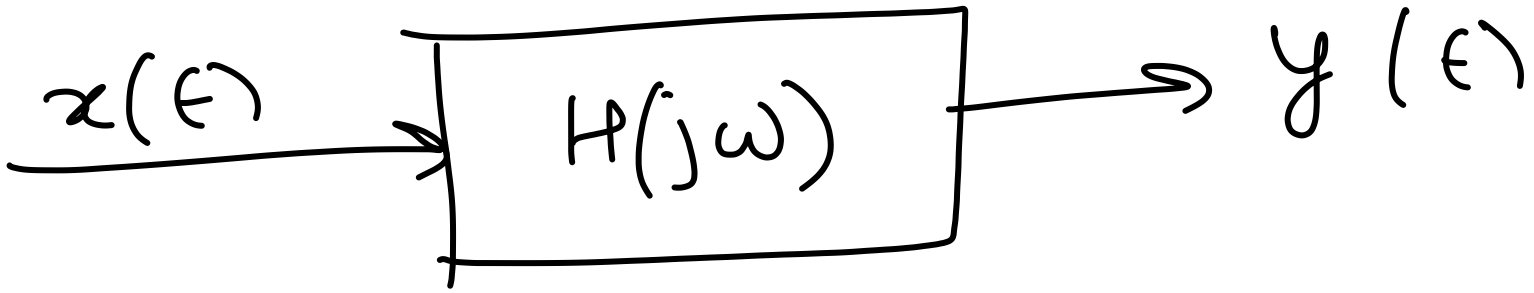
$$= \frac{A}{\omega^2 + B^2} \sqrt{B^2 + \omega^2}$$

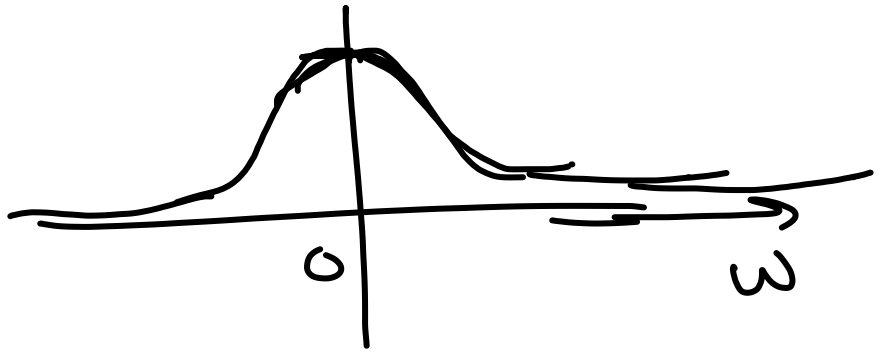
$$= \frac{A}{\sqrt{B^2 + \omega^2}}$$

$$= \frac{1/4}{\sqrt{\omega^2 + \frac{(3-\omega^2)^2}{16}}}$$

$$= \frac{1/4 \times 1}{\sqrt{16\omega^2 - \omega^4 - 6\omega^2 + 9}}$$

$$= \frac{1}{\sqrt{10\omega^2 - \omega^4 + 9}}$$



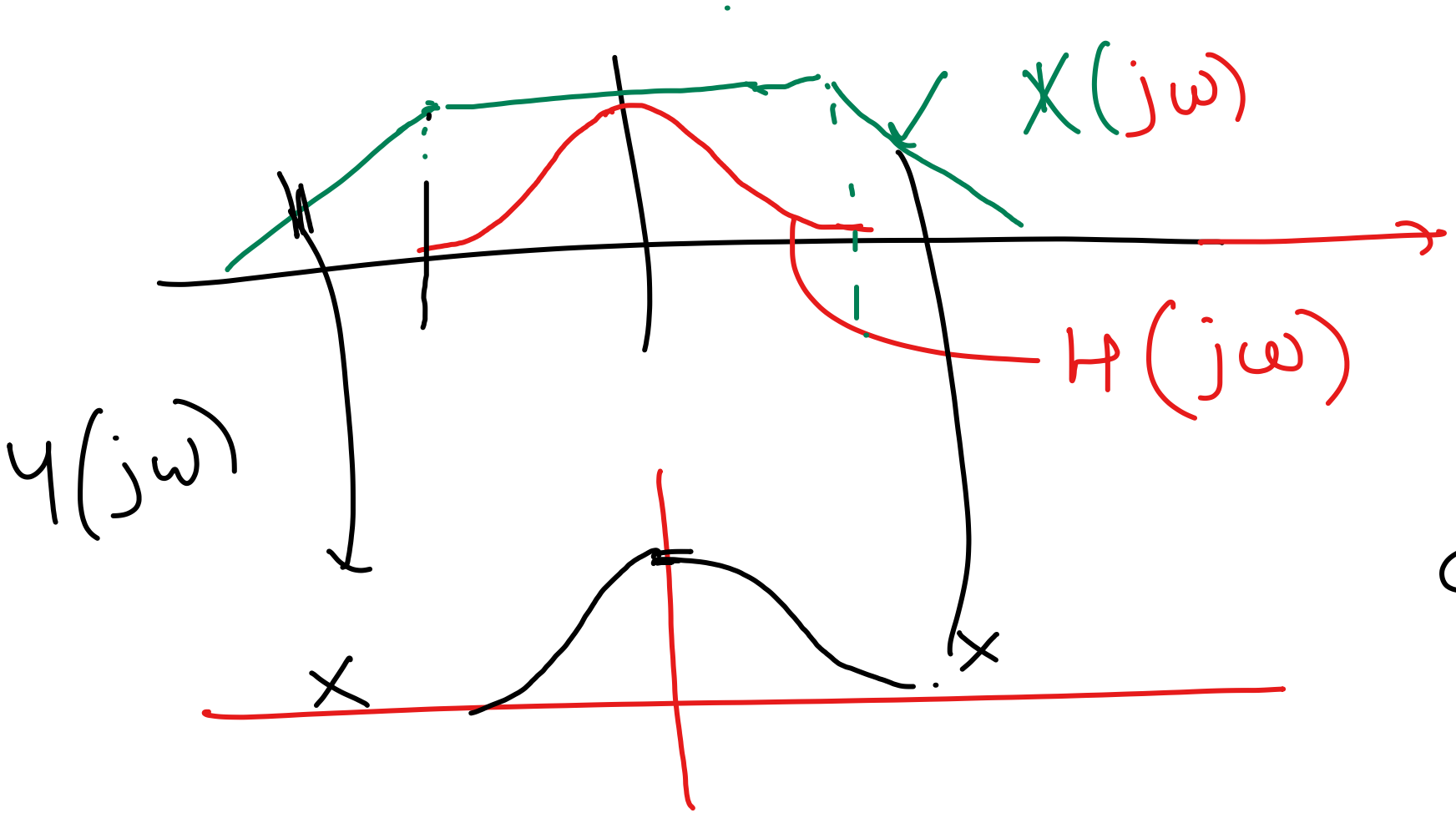


$$Y(j\omega) = X(j\omega)\,H(j\omega) =$$

$$\frac{X(j\omega)}{3}$$

at $\omega \to 0$ $\quad Y(j\omega) =$

at $\omega \to \infty$ $\quad Y(j\omega) = 0$



only low frequencies of x(t) are passed to the o/p

(b) $H_2(s) = \dfrac{s}{s^2+s+1}$ $\quad Re\{s\} > -\frac{1}{2}$

$j\omega$ - axis is included in ROC

F.T. exists

$$H(j\omega) = \frac{j\omega}{j\omega - \omega^2 + 1}$$

$\omega \to 0 \quad |H(j\omega)| \to 0$

$\omega \to \infty \quad |H(j\omega)| \to 0$

divide by ω^2 on Numerator & denom

$$H(j\omega) = \frac{\frac{1}{\omega}}{\frac{1}{\omega} - 1 + \frac{1}{\omega^2}} =$$

Band pass $\times$

(3) $H_3(s) = \frac{s^2}{s^2+2s+1}$ $\quad \mathrm{Re}\{s\} > -1$

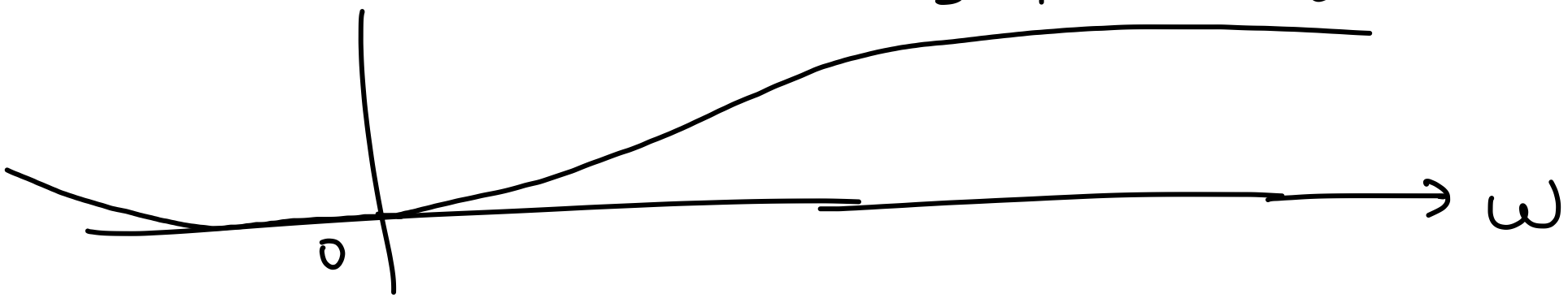
F.T. exists

$$H_3(j\omega) = \frac{\omega^2}{\omega^2 - 2j\omega - 1}$$

$\omega \longrightarrow 0$ $\quad H_3(j\omega) \longrightarrow 0$

$\omega \longrightarrow \infty$ $\quad |H_3(j\omega)| = 1$

Highpass filter.



Z-transform

Consider an LTI system with impulse response $h[n]$

$h[n] \xrightleftharpoons{\text{Z-transform}} H(z)$ with certain ROC

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

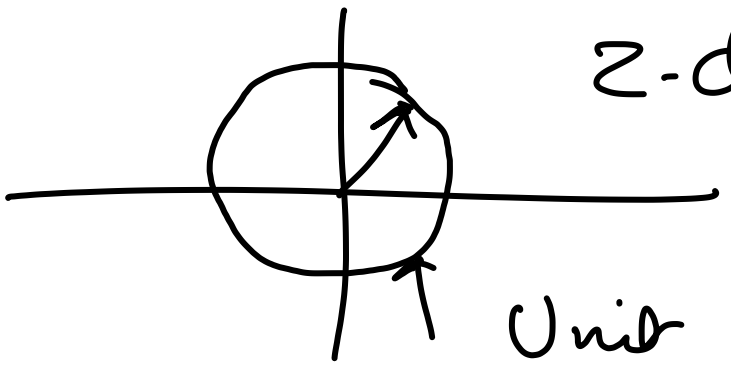
Let $z = e^{j\omega} = |z| e^{j\omega}$ $|z| = 1$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n}$$

D.T.F.T.

on unit circle,

is my D.T.F.T.



In general $Z = |Z| e^{j\omega}$ = polar representation of a complex no.

$= r e^{j\omega}$

radius in the Z-domain.

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} h[n] (re^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left(h[n] r^{-n} \right) e^{-j\omega n}$$

$H(z) \equiv$ F.T. of $\{ h[n] r^{-n} \}$

r belongs to my ROC.

$H(z)$ $\equiv$ region of convergence

Stable System $\Rightarrow$ satisfies BIBO condition

$\Downarrow$

$\sum |h[n]| < \infty$ system is absolutely Sumable

$\Downarrow$

D.T.F.T. of $h[n]$ exists.

$\Downarrow$

Unit circle is included in

ROC of the Z-transform of $h[n]$

& Vice-versa

$\Downarrow$

if ROC of Z-transform of $h[n]$ includes Unit circle,

the corresponding LTI system is a stable system.

Causality -

if $H(z)$ is a rational fn. in z

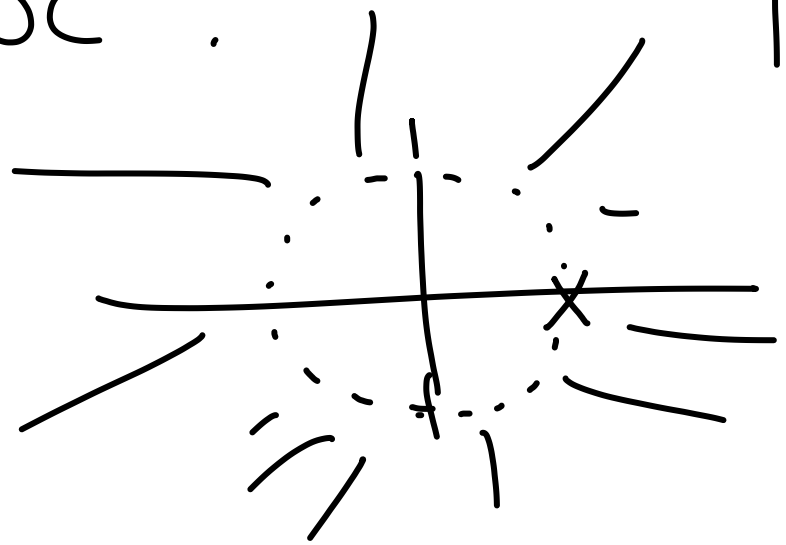
$$\Rightarrow H(z) = \frac{N(z)}{D(z)} = \frac{\text{polynomial in } z \text{ or } z^{-1}}{\text{polynomial in } z \text{ or } z^{-1}}$$

& ROC is outside the outermost pole

then, the system is causal.

if $h[n] = a^n u[n] = \frac{1}{1-az^{-1}}$

ROC : $|z| > |a|$



z-domain

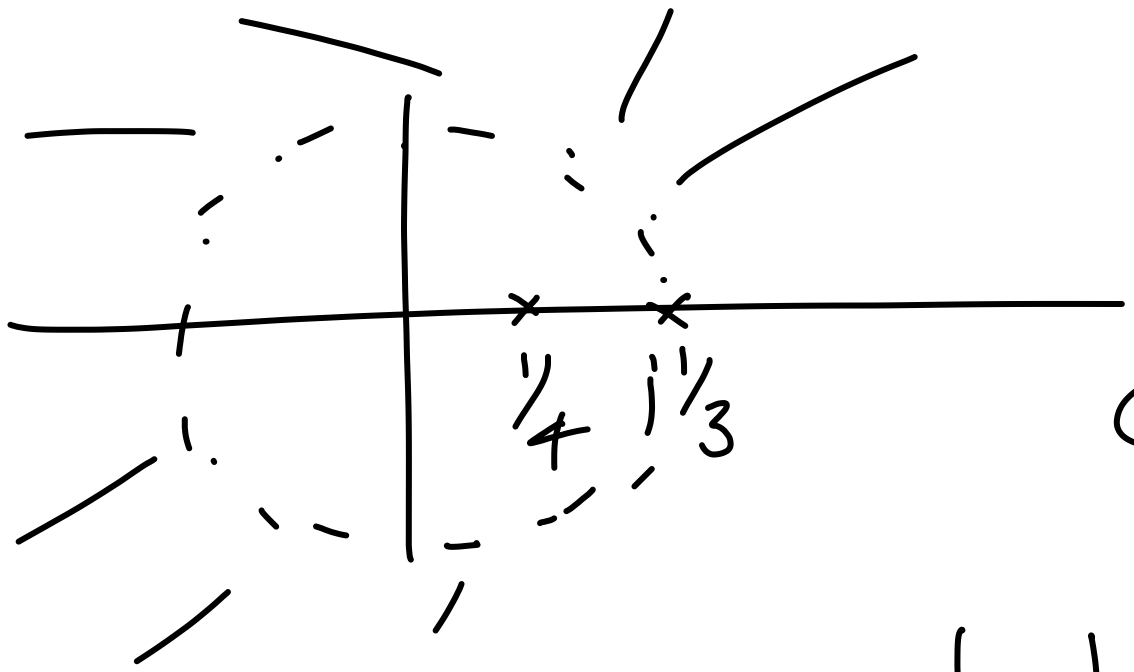
$$h[n] = a^n u[n] + b^n u[n]$$

$$H(z) = \frac{1}{1 - az^{-1}} + \frac{1}{1 - bz^{-1}}$$

$\Downarrow$ $|z| > |a|$

$\Downarrow$ $|z| > |b|$

Let $a = 1/3$, $b = 1/4$



ROC :

Common region of intersection

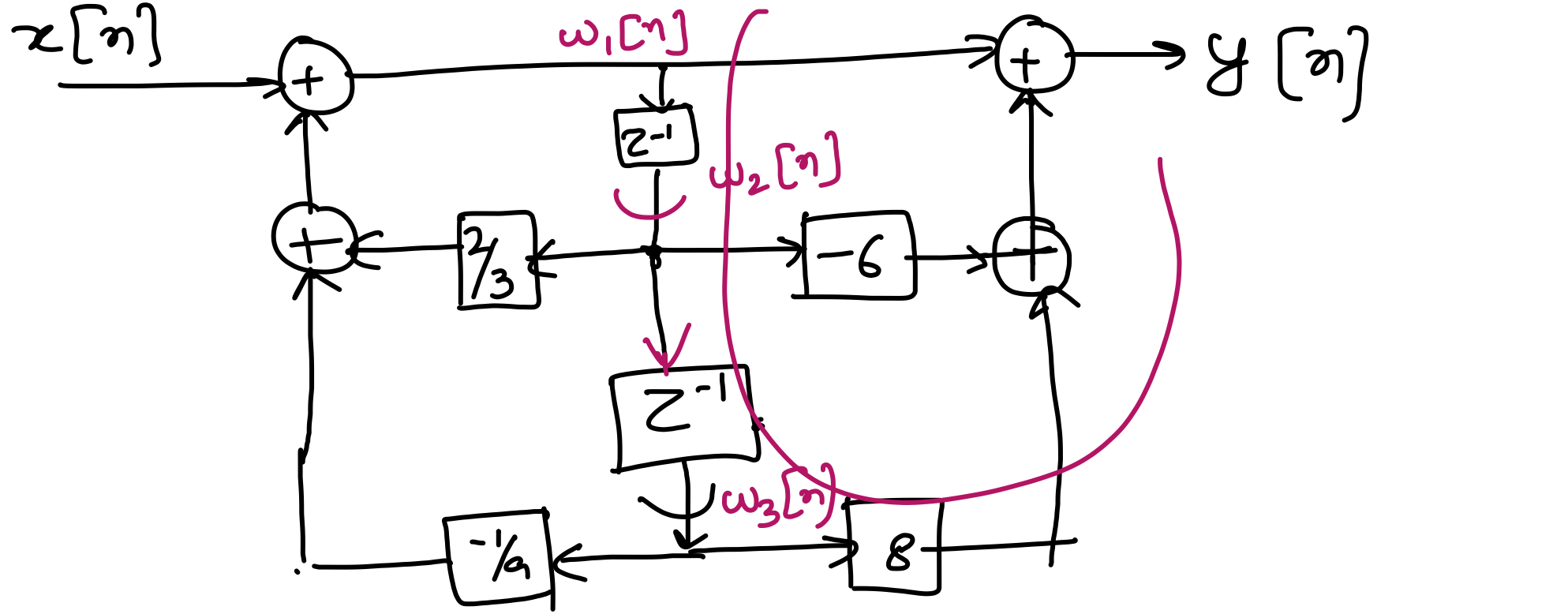
$|z| > 1/3$

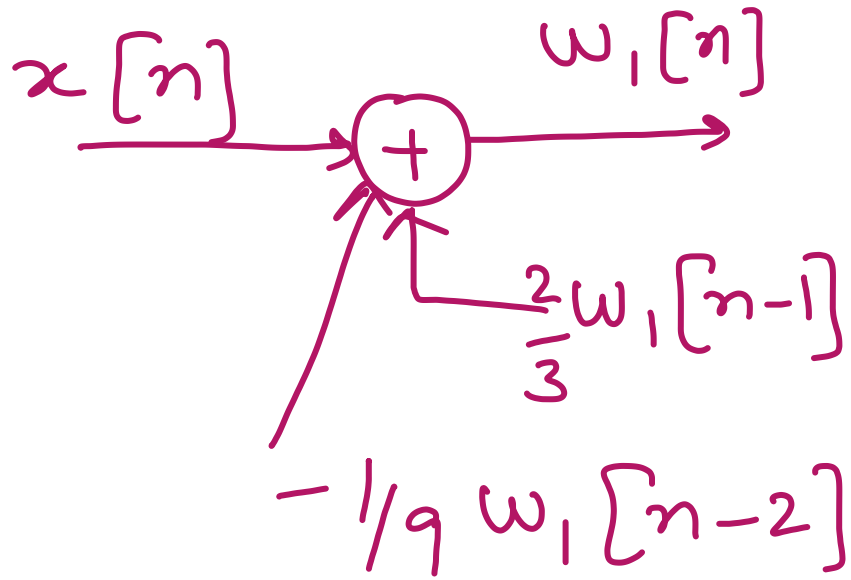
if an LTI system is causal, $\{h[n] = 0 \; n<0\}$

ROC is outside the outermost pole.

Converse is true, if $H(z)$ is a rational fn. in z.

10·18





$$W_1(z) = X(z) + \frac{2}{3} z^{-1} W_1[z] - \frac{1}{9} z^{-2} W_1[z]$$

$$W_2[z] = z^{-1} W_1[z]$$

$$W_3[z] = z^{-1} W_2[z] = z^{-2} W_1[z]$$

$$g[n] \rightleftharpoons G(z)$$

$$y[n] = g[n-1] \rightleftharpoons ?$$

$$G(z) = \sum_{n=-\infty}^{\infty} g[n]\, z^{-n}$$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} g[n-1]\, z^{-n}$$

$m = n-1$

$$= \sum_{m=-\infty}^{\infty} g[m]\, z^{-(m+1)}$$

$$= z^{-1} \sum_{m=-\infty}^{\infty} g[m]\, z^{-m}$$

$$= z^{-1} G(z)$$

$$g[n] \rightleftharpoons G(z) \qquad \text{ROC}: R$$

$$g[n-1] \rightleftharpoons z^{-1} G(z) \quad : \quad R - \{z=0\}$$

$$\frac{W_1(z)}{X(z)} = \frac{1}{1 - \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2}}$$

$$Y(z) = W_1(z) - 6z^{-1}W_1(z) + 8z^{-2}W_1(z)$$

$$\frac{Y(z)}{W_1(z)} = 1 - 6z^{-1} + 8z^{-2}$$

$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{W_1(z)} \times \frac{W_1(z)}{X(z)} = \frac{1 - 6z^{-1} + 8z^{-2}}{1 - \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2}}$$

$$= \frac{N(z)}{D(z)}$$

Roots where

$N(z) = 0 \Rightarrow$ zeros of system

Roots where

$D(z) = 0 \Rightarrow$ poles of the system

$1 - \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2} = \left(1 - \frac{1}{3}z^{-1}\right)^2 \rightarrow$ repeated

$\therefore$ causal LTI

$\rightarrow$ system is stable.